HAILO
Empowering Intelligence

Hailo-15H SBC 2.x
Single Board Computer Kit
Quick Start Guide

Revision 1.2

March 2026

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This version of the document supersedes all previous versions.

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Revision History

This section describes top-level differences in the versions of this document.

Version	Date	Description
0.1	December 2023	Initial release
0.2	February 2024	Added kit component picture Updated SPI flash script command
0.3	March 2024	Installation details for USB to UART Bridge Changed order of instructions to program SPI flash first Updated demo pipeline to latest release
0.4	June 2024	Version review and refresh Added Sensor configuration instructions. New demo application image Updated SPI flash programming command
0.5	September 2024	Clarified details for downloading the .wic file
0.6	June 2025	Corrected CR error in SPI flash programming Updated trademark information
0.7	October 2025	Updated instructions for software download and installation for Hailo-15H SBC 2.x version
1.0	February 2026	Added diagrams to assist in identifying GPIO pins 14. 16 and 18 for UART1 adapter board connection. Simplified the SPI_flash_program command in Section 2.3.3 Updated Section **Error! Reference source not found.** Running the Camera Viewer
1.1	February 2026	Align with Vision Processor Software Package version 1.10.1
1.2	March 2026	Align with Vision Processor Software Package version 1.11.0

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1. Overview

1.1. Introduction

This guide provides an overview of the Hailo-15H SBC 2.x, a comprehensive platform for developing and prototyping smart cameras powered by the Hailo-15H AI vision processor. Its powerful AI capabilities enable both AI-powered image enhancement and efficient, full-scale processing of multiple complex deep learning applications. For a complete description of the Hailo-15 features and capabilities, refer to the **Hailo-15 Feature-list and capabilities**[1] document.

The Hailo-15H SBC 2.x showcases high-quality video streaming and advanced AI inference capabilities, including:

- Direct connection to supported camera sensors.
- Versatile interfaces for connecting, testing, and power measurement
- Easy integration with existing customers' platforms.
- Seamless integration with popular software frameworks.

The Hailo-15H SBC 2.x contains:

- Single Board Computer (SBC) featuring the Hailo-15H VPU.
- 12V DC power supply, microSD™ card, spacers, heatsink for the Hailo-15H VPU, USB to UART bridge, USB-to-UART cable.
- Hailo Vision Processor Software Package

This guide is intended to help users efficiently utilize the Hailo-15H SBC.

It provides essential hardware setup and software configuration steps, as well as instructions for executing the Hailo-15H demo application.

Note:

All photos and descriptions in this guide refer to Hailo-15H SBC 2.x. kit with camera module based on the IMX678 sensor.

The camera sensor module and accompanying cable are not included in the Hailo-15H SBC 2.x and must be ordered separately from Hailo. See Section 5.1 for further information on available camera sensor modules.

The steps described in this guide were verified using Hailo-15H SBC 2.x. kit, camera module with IMX678 camera sensor and Intel x86-64 based PC with Ubuntu 22.04 LTS, Python 3.8.10 and 3.10.12 and pyspiflash 0.6.3.

[1] Available in the Hailo Developer Zone (Link)

2. Hardware and Software Setup

This section details the hardware and software configuration steps required for use with Hailo-15H SBC Revision 2.x

2.1. Prerequisites

The following components* are required in addition to the Hailo-15H SBC 2.x:

- Ethernet cable RJ45 CAT5 or above.
- X86-64 based computer running Ubuntu® 22.04 ("PC").

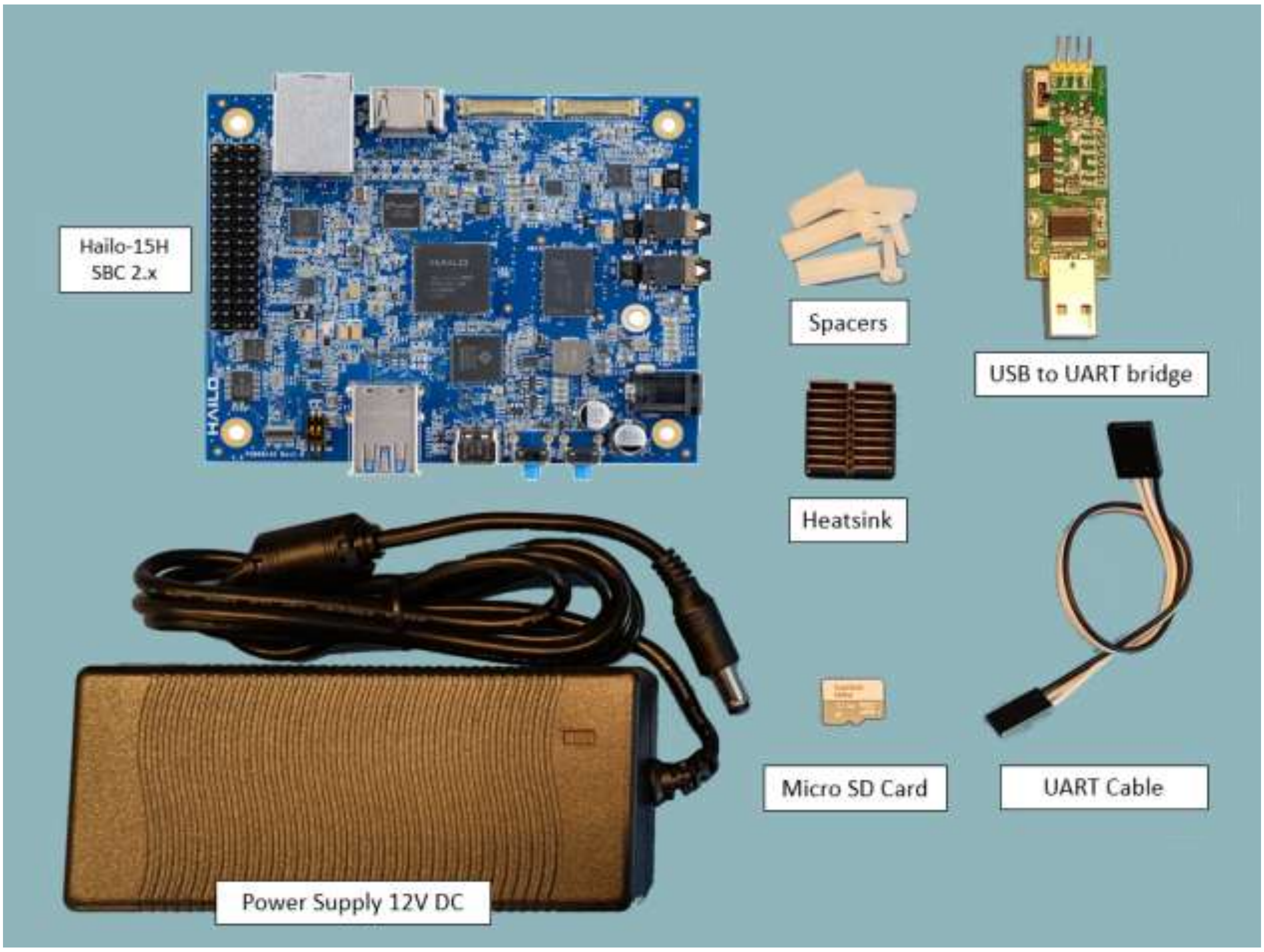


Figure 1. Hailo-15H SBC 2.x components

*See Table 1 for the complete list of the components included in the Hailo-15H SBC 2.x kit

2.2. Hailo-15H SBC Key Components

Figure 2 and 3, below display the front and rear views of the Hailo-15H SBC 2.x, highlighting the key component names.

Note that the Hailo-15H SBC 2.x R includes a 4GB LPDDR4x device.

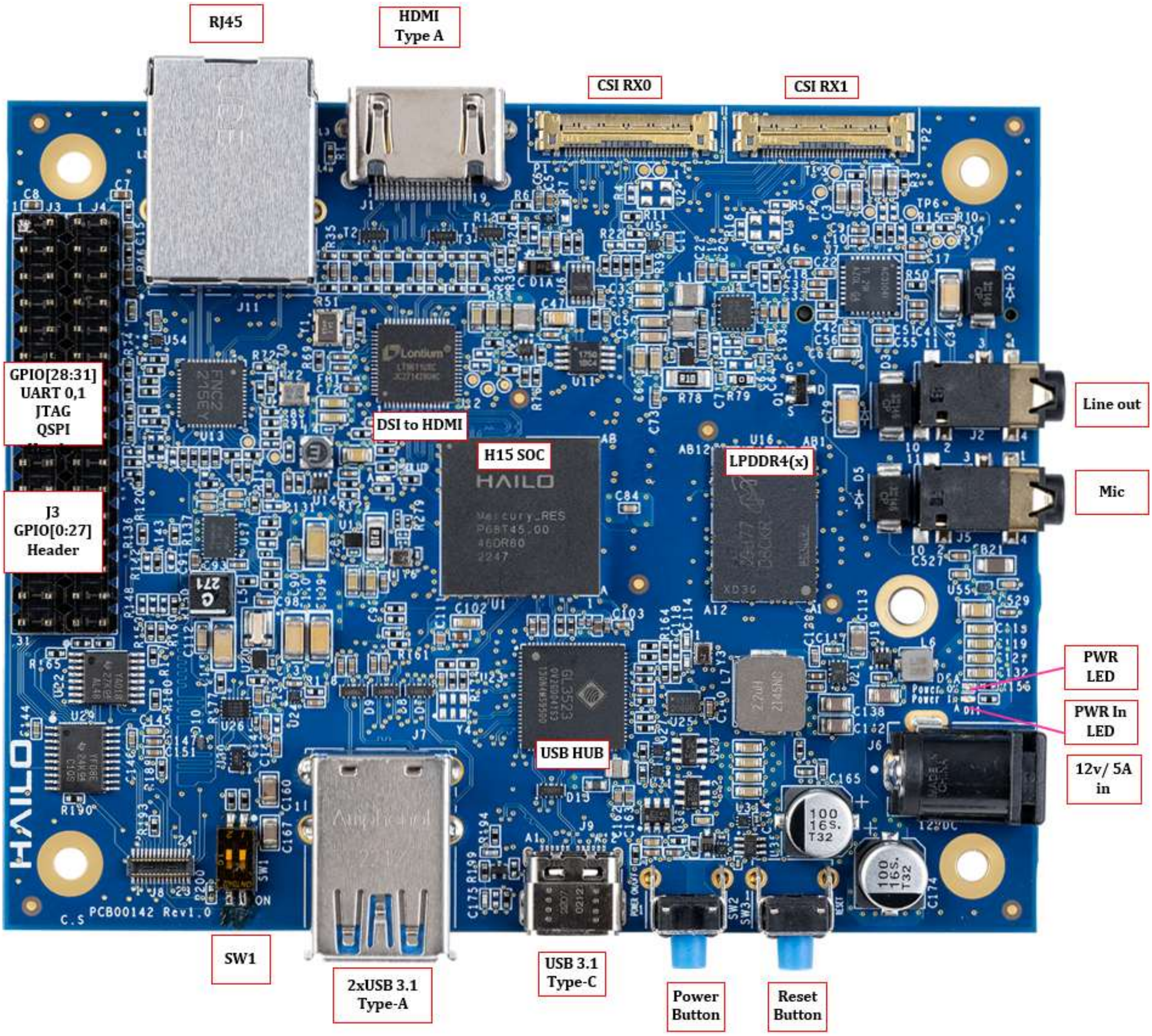


Figure 2. Hailo-15H SBC front view

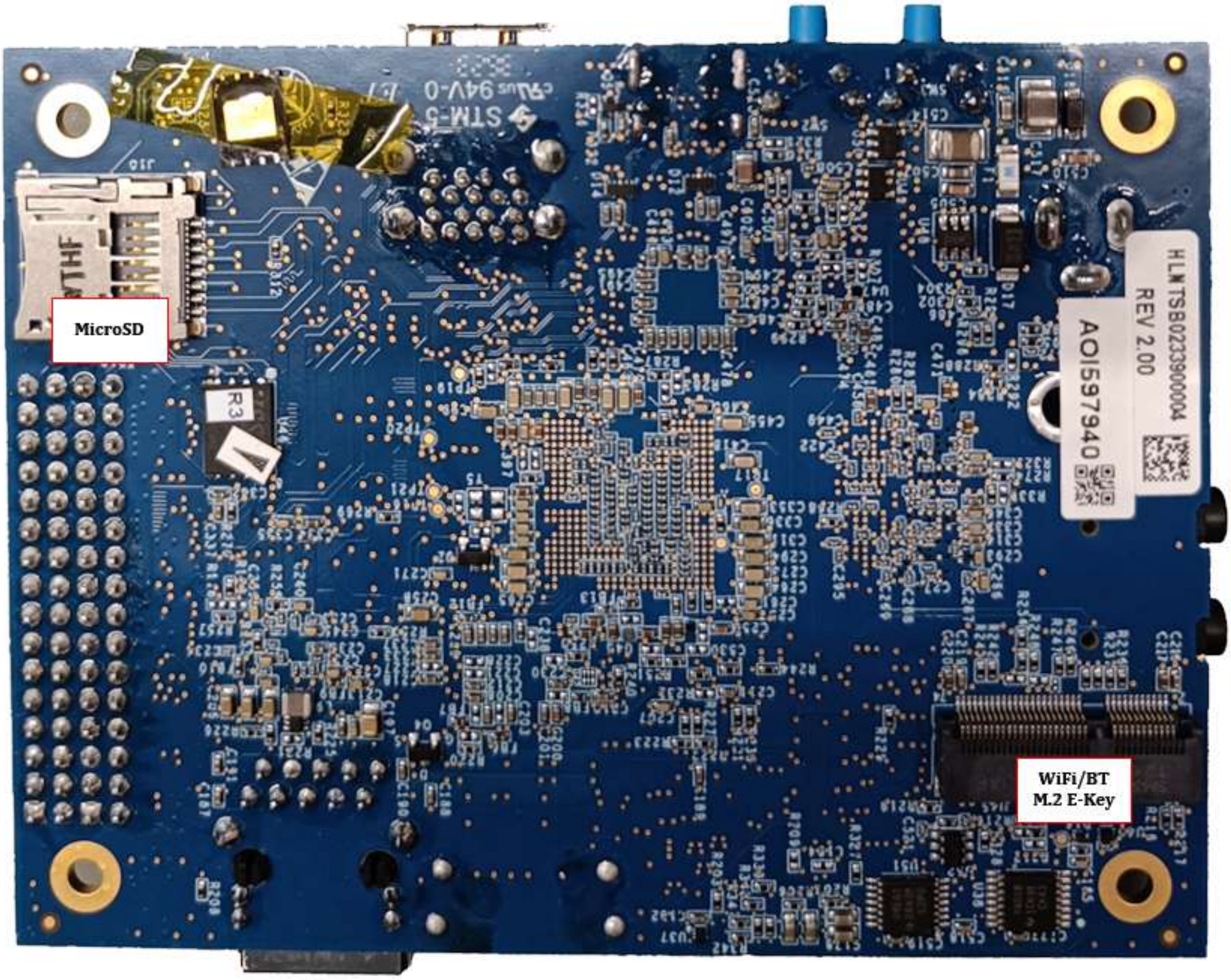


Figure 3. Hailo-15H SBC rear view

2.3. Hailo-15H SBC 2.x Software Image

The Hailo-15H SBC2.x includes an SD card preloaded with the Hailo **Vision Processor Software** image.

Follow the steps below to verify that the Hailo-15H SBC is loaded with the latest Hailo-15H **software version**.

1. Refer to the Hailo Vision Processor Software Package Release Notes, available in Hailo Developer Zone Link to identify the most up-to-date version of the Hailo-15H software.
2. The Hailo-15H SBC is statically configured with IP address 10.0.0.1 on subnet 10.0.0.0/24. The development host must be assigned an IP address within the same subnet e.g., 10.0.0.2.
3. Login to the Hailo-15H SBC 2.x kit via SSH using the terminal command:

`ssh root@10.0.0.1`

(The default password is 'root')

4. Enter the command `cat /etc/os-release`

If the Hailo-15H SBC 2.x has been loaded with software version **1.11.0**, continue directly to Section 2.4; Otherwise, follow the steps detailed in Section 2.3.1 to update the latest software version on the Hailo-15H SBC.

2.3.1. Hailo-15H SBC 2.x Software Image Update

The Hailo-15H SBC software image update includes the following steps:

- Configure the Hailo-15H SBC 2.x to boot from UART (Section 2.3.2).
- Program the SPI flash with the required bootloader and firmware images (Section 2.3.3). This step utilizes a recovery mechanism implemented in the Hailo-15 boot ROM.
- Reconfigure the SBC to boot from SPI flash (Section 2.3.4).
- Program the software image to the SD card (Section 2.3.5).

After the update has completed, the SBC automatically reboots and runs the updated Hailo Vision Processor Software image.

⚠ WARNING ⚠

Follow the steps detailed in this guide exactly as they are described and in the correct order, failure to do so may damage the hardware.

2.3.2. Setting the Hailo-15H SBC 2.x for Boot from UART

The Hailo-15H SBC has a recovery mechanism implemented in the Boot ROM. It can be used to program a blank or corrupted SPI flash. This method utilizes physical access to UART1, and its configuration is detailed below.

1. Power off the Hailo-15H SBC. See Figure 4

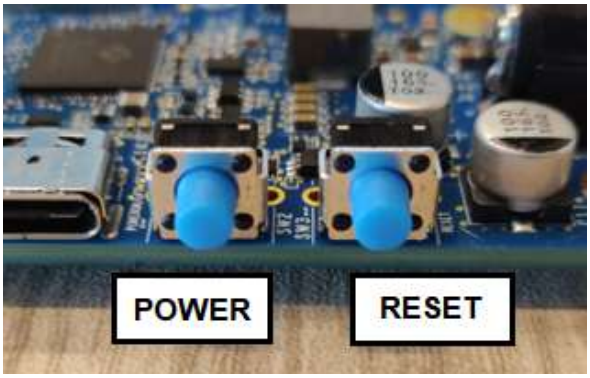


Figure 4. Power and reset buttons in the Hailo-15H SBC2.x

2. If present using fine tweezers, remove the orange tape covering the DIP switches.
3. Using fine tweezers, set the bootstraps to boot from UART by setting the DIP switches as shown in Figure 5
SW1: 1=ON , SW1: 2=OFF.

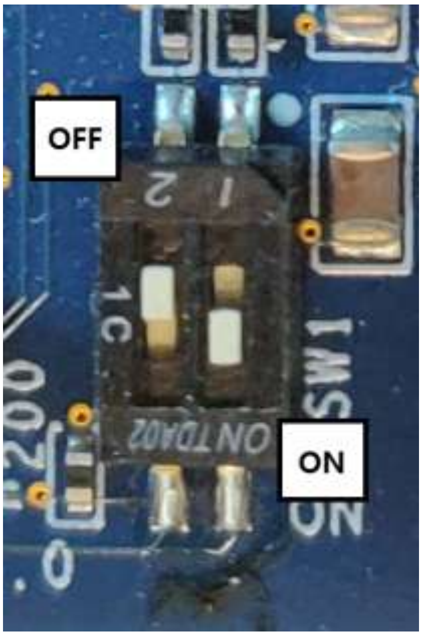
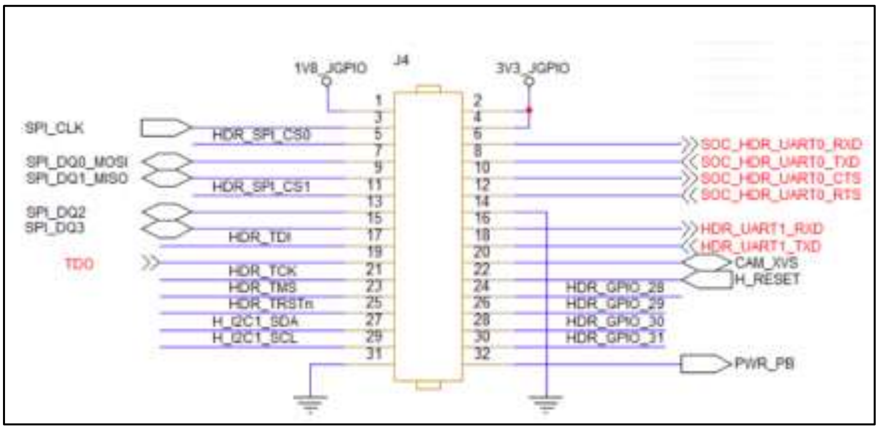


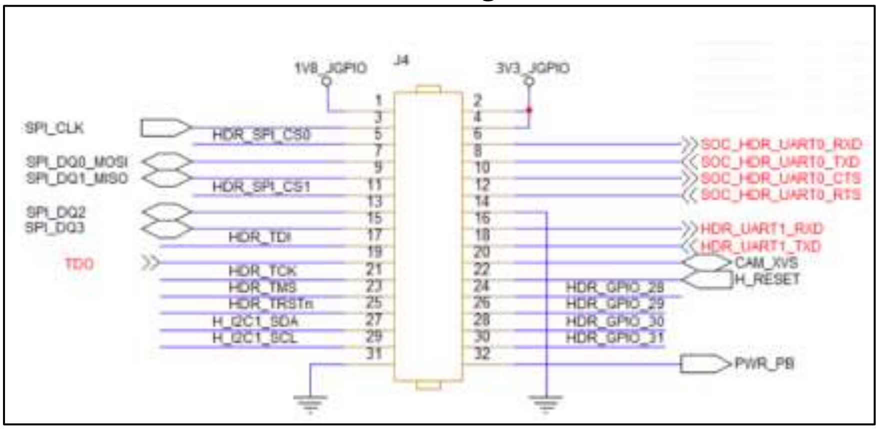
Figure 5. Setting the DIP switches SW1 and SW2

2.3.2.1. Preparation and Connection of the UART Adapter Board

1. Before starting, the next steps, confirm on the adapter board that the jumper is attached to the 1.8[v], as displayed in the pink rectangle in Figure 8. In a case where a different UART adapter is used, find the voltage control and set it to 1.8V. **Note: setting the wrong voltage will cause damage to the device**.
2. Verify the cable connections on the UART adapter side - make sure only the GND, Rx & Tx ports are connected. **Note: connecting Vcc will cause damage to the device**.



With the aid of Figure 6 and



3. Figure 7 below, identify on the J4 header the GPIO pins 14, 16 and 18.

.

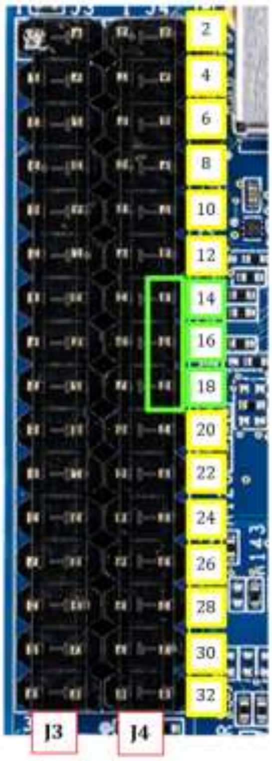


Figure 6. Locations of GPIO pins 14, 16 and 18 on the J4 header

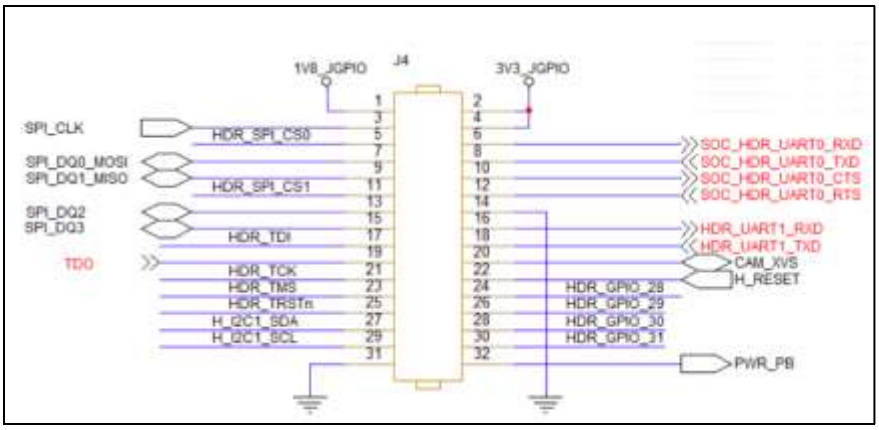


Figure 7. Schematic diagram of the Header J4 and GPIO pin configurations

4. Insert the cable connections on the SBC side (J4 header) See Figure 8.
5. Make sure:
    - Header pin 14 (black in the image below) is connected to GND,
    - Header pin 16 (SBC Rx) (green in the image below) is connected to UART Adapter's Tx
    - Header pin 18 (orange in the image below) (SBC Tx) is connected to UART Adapter's Rx.

Note: the cable color coding may be changed in future kits.

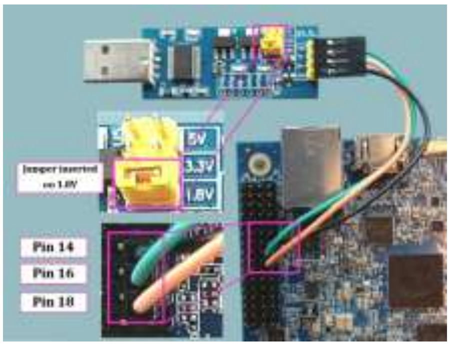


Figure 8. UART1 adapter board and cable (jumper and pin connections magnified)

6. Using the buttons shown in Figure 9, power on the SBC.
7. Reset the SBC

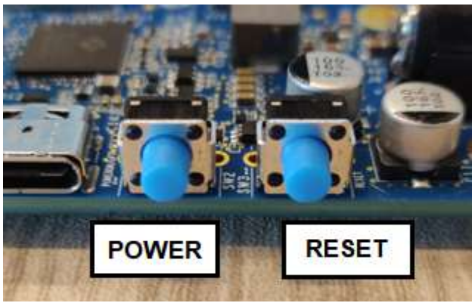


Figure 9. Power and reset buttons

2.3.3. Programming the U-Boot to SPI Flash

1. Download the Hailo-15H Vision Processor Software Package from Hailo Developer Zone (See Link) as shown in Figure 10

Package name: Hailo-15H Vision Processor Software Package SBC 2.X

File name: *hailo_vision_processor_sw_package_1.11.0_hailo15h_SBC_2.X.tar.gz*

Note that the **Hailo-15 vision processor software package release notes** document provides briefing for the Vision Processor software package. For more details on its content, structure, key components and latest features can be found in the Documents section, accessed via the Documentation tab in the Developer Zone.

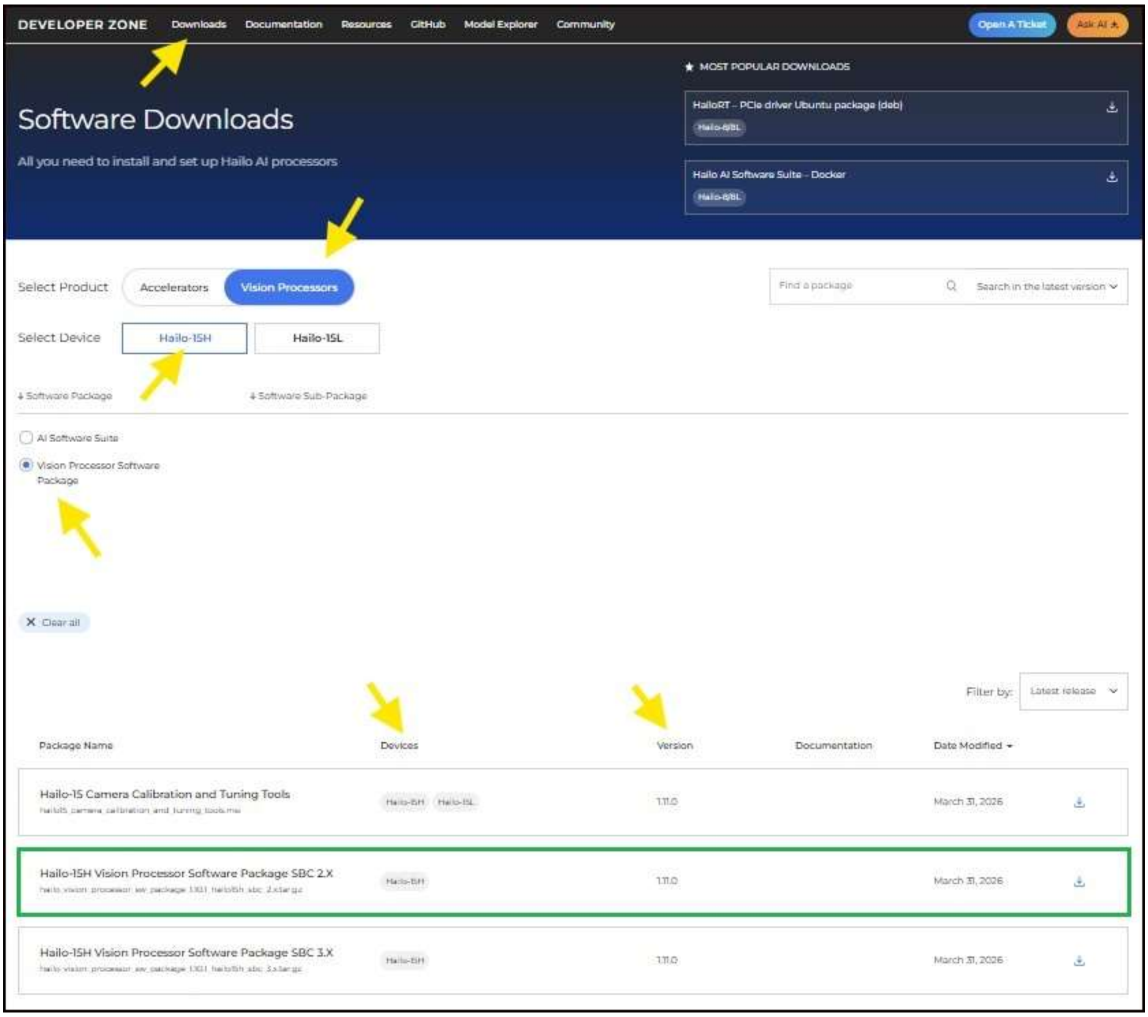


Figure 10. Software Download Page in the Hailo Developer Zone

2. On the development host terminal, run the following commands:

```
pip install tools/hailo15_board_tools-1.11.0-py3-none-any.whl
sudo apt-get install u-boot-tools
```

Note:

- The .whl file is part of the Hailo *Vision Processor Software Package* under the tools folder.

3. On a development host terminal
   a. Edit (or if it does not exist, create) the following file: */etc/udev/rules.d/99-usb.rules* and add the following line:

```
SUBSYSTEM=="usb", ATTR{idVendor}=="0403", ATTR{idProduct}=="6015",
GROUP="plugdev", MODE="0664"
```

   b. Run the following commands:

```
sudo udevadm control --reload
```



```
sudo udevadm trigger
sudo adduser $USER plugdev
```

   c. Reboot the development host machine

4. On the development host terminal, load the recovery firmware using the following command:

```
uart_boot_fw_loader \
--serial-device-name /dev/ttyUSB0 \
--firmware ./prebuilt/sbc/hailo15_uart_recovery_fw.bin
```

Note that the .bin file is part of the *Hailo-15 Vision Processor Software Package* and is located under the *prebuilt/sbc* folder

5. On the development host terminal, load the SPI flash images using the commands below.

Note that the files mentioned in the hailo15_spi_flash_program command below are part of the Hailo *Vision Processor Software Package* and are located under the *prebuilt/sbc* folder

```
cd prebuilt/sbc

hailo15_spi_flash_program \
--scu-bootloader ./hailo15_scu_bl.bin \
--scu-bootloader-config ./scu_bl_cfg_a.bin \
--scu-firmware ./hailo15_scu_fw.bin \
--uboot-device-tree ./u-boot.dtb.signed \
--bootloader ./u-boot-spl.bin \
--bootloader-env ./u-boot-initial-env \
--customer-certificate ./customer_certificate.bin \
--uboot-tfa ./u-boot-tfa.itb \
--uart-load \
--serial-device-name /dev/ttyUSB0
```

2.3.4. Boot the U-Boot Secondary Program Loader from SPI Flash

1. Power off the Hailo-15H SBC 2.x.
2. Using fine tweezers, set the bootstraps to boot from SPI Flash by setting the DIP switches to OFF as shown in Figure 11.

   SW1: 1=OFF   SW1.2=OFF

Figure 11. Setting DIP switches SW1 and SW2 to OFF

3. Press on the power button to power up the Hailo-15H SBC 2.x See Figure 12 for the location of the button.
4. Press on the reset button to reset the Hailo-15H SBC 2.x.

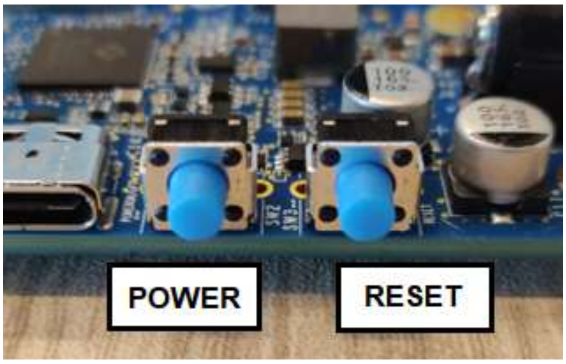


Figure 12. Hailo-15H SBC2.x power and reset buttons

2.3.5. Programming the Image to the SD Card

The following sections describe how to program a pre-compiled image to the SD card.

2.3.5.1. Ethernet Cable Connection

Connect the Ethernet cable to the development host and the Hailo-15H SBC 2.x RJ45 connector as shown in Figure 13

Figure 13. Ethernet RJ45 connector

2.3.5.2. Setup TFTP Server on Host

1. Install the TFTP server:

```
sudo apt updated
sudo apt install tftpd-hpa
```

2. Edit the following file */etc/default/tftpd-hpa* and add the following lines:

```
TFTP_USERNAME="tftp"
TFTP_DIRECTORY="/var/lib/tftpboot"
TFTP_ADDRESS="0.0.0.0:69"
TFTP_OPTIONS="--secure"
```

3. Create the TFTP directory:

```
sudo mkdir -p /var/lib/tftpboot
sudo chmod -R 777 /var/lib/tftpboot
```

4. Restart the TFTP server:

```
sudo systemctl restart tftpd-hpa
```

2.3.5.3. Programming the SD Card

1. The Hailo-15H SBC is statically configured with IP address 10.0.0.1 on subnet 10.0.0.0/24. The development host must be assigned an IP address within the same subnet e.g., 10.0.0.2
2. Ensure the dip switch is set to OFF.
   SW1: 1=OFF SW1.2=OFF
3. Copy the FIT image file into the TFTP directory:

```
sudo cp fitImage /var/lib/tftpboot/
```

4. Copy the swupdate files into the TFTP directory:

```
sudo cp swupdate-image-hailo15-sbc.ext4.gz /var/lib/tftpboot/
sudo cp hailo-update-image-hailo15-sbc.swu /var/lib/tftpboot/
```

5. Ensure that the USB-to micro-USB cable between the host and the Hailo-15H SBC 2.x is connected and connect serially by entering the command:

```
sudo picocom -b 115200 /dev/ttyUSB0
```

6. Reset the Hailo-15H SBC 2.x, with the reset button. See Figure 14
7. From the U-Boot menu select “SD Card Board Init” as shown in Figure 14.

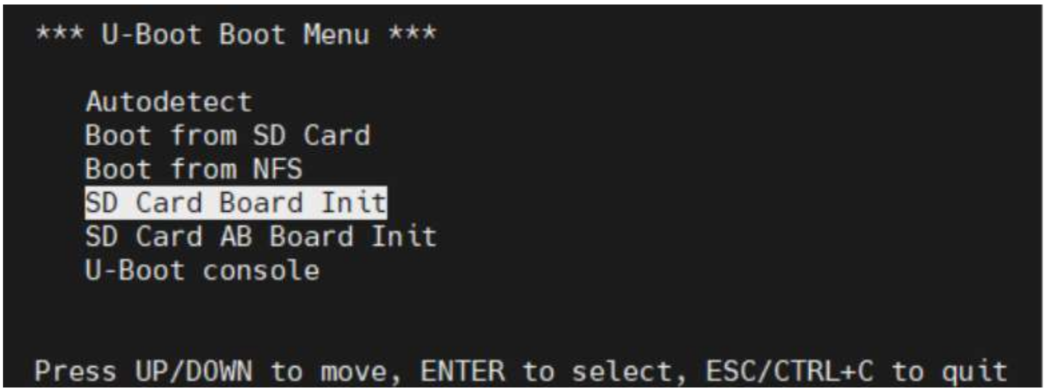


Figure 14. U-Boot menu SD card

8. U-Boot will now download the SW update and begin the update procedure.
9. Wait for the procedure to be completed by observing the traces below, followed by an automatic device reboot as shown in Figure 15.

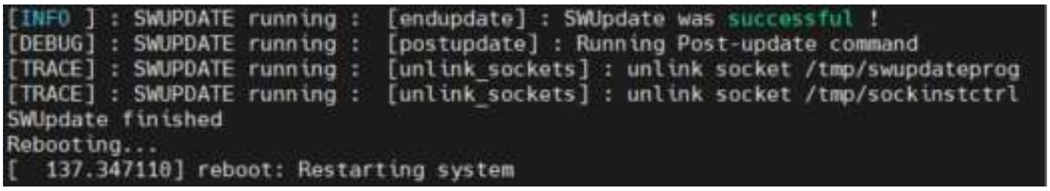


Figure 15. Software update procedure completion

2.4. Ethernet Cable Connection

If an Ethernet cable has not yet been connected, connect it to the development host and the Hailo-15H SBC's RJ45 connector as shown in Figure 16

Figure 16. Ethernet RJ45 connector

2.5. Camera Sensor Module Connection

It is recommended to use the IMX678 Camera Sensor Module.

1. Confirm the Micro SD card is located inside the Hailo-15H SBC Micro SD card slot
2. Confirm the SW1 DIP switches are set OFF as shown in Figure 17.

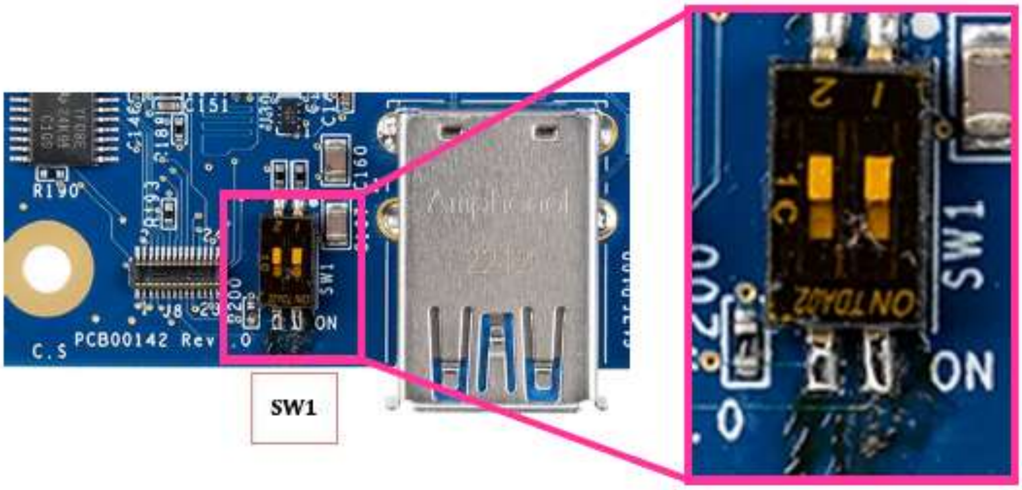


Figure 17. Hailo-15H SBC 2.x DIP SW 1

3. Connect one end of the camera sensor module cable to the IMX 678 camera sensor module as shown in Figure 18.

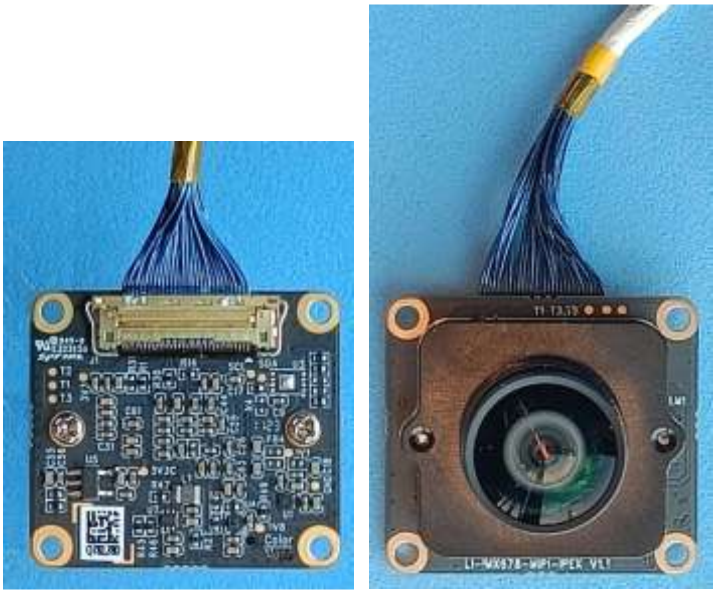


Figure 18. Sensor cable connection to the IMX 678 camera sensor module

4. Connect the other end of the camera sensor module cable to the MIPI-CSI2 connector labeled ‘P2’ on the Hailo-15H SBC 2.x as displayed in Figure 19.

Figure 19. Sensor cable connection to camera sensor module

2.6. Powering up the Hailo-15H SBC

1. Press the button labeled ‘PWR’ to power up the Hailo-15H SBC, the adjacent green LED will light ON. See Figure 20 and Figure 21.
2. When powering on the Hailo-15H SBC 2.x (OFF → ON), a short press will turn the Hailo-15H SBC 2.x ON.

NOTE: When powering off the Hailo-15H SBC 2.x (ON → OFF), a long press will turn the Hailo-15H SBC OFF.

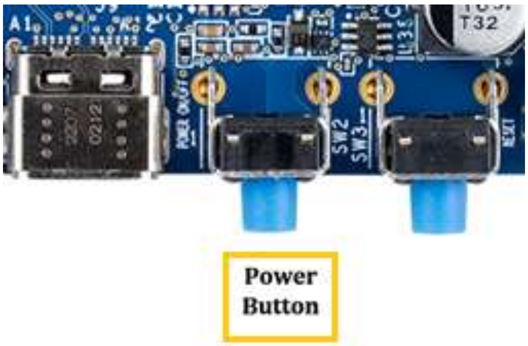


Figure 20. Power Button

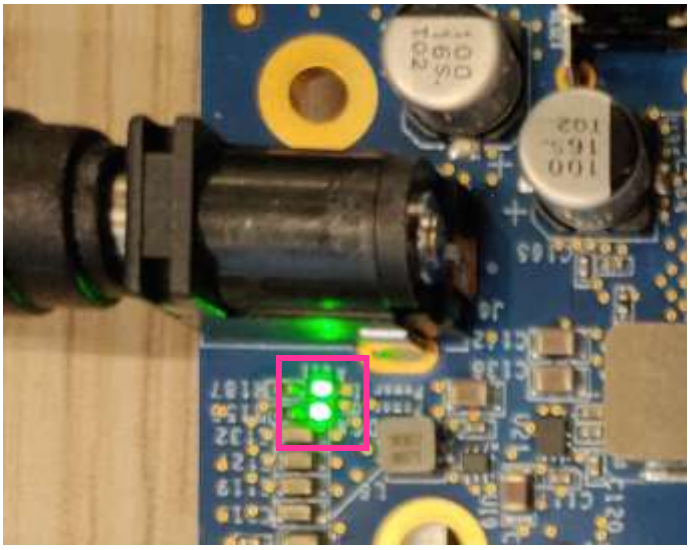


Figure 21. Adjacent green LEDs lighting up

3. Runing Hailo Camera Viewer

The Hailo Camera Viewer is a video streaming interface allowing control of standard camera setup parameters such as video resolution, FPS, imaging profiles and preferences, encoding and overlay.

HDR and low-light imaging modes as well as a basic detection pipeline can be operated through functions in the AI & Image Enhancement tabs.

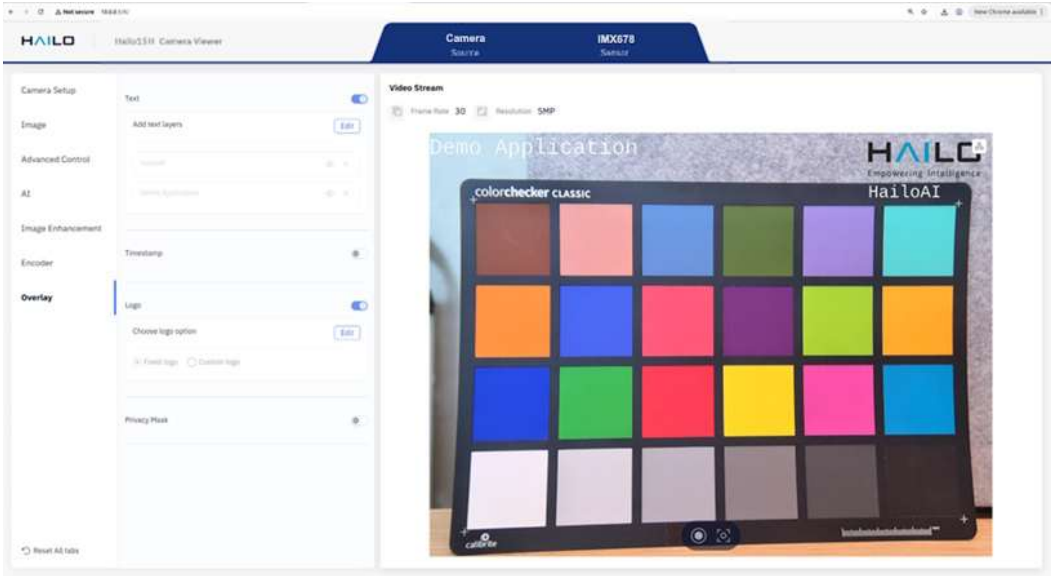


Figure 22. Camera viewer

3.1. Launch the Hailo Camera Viewer

Open a terminal on the host PC and enter the following commands:

1. Connect via SSH:

   `ssh root@10.0.0.1`

Note: the default password: root

2. Configure the Camera Sensor

   `setup_hailo_sensor.sh`

3. Launch the camera server:

   `camera-viewer-server`

Leave the camera-viewer-server running; it serves the **Camera Viewer** web UI.

4. Open a browser and navigate to:

   `http://10.0.0.1/#/`

3.2. Live Video Streaming

The Camera Viewer can be used to demonstrate full pipeline, from video capture through ISP, encoding, streaming, and display.

- In Camera Viewer, click **Play** to start the live stream.

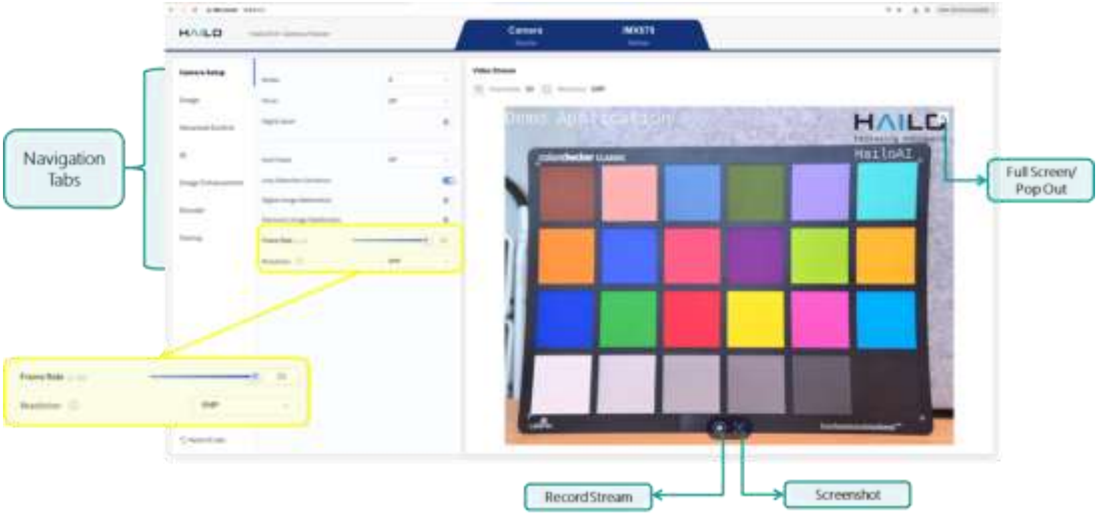


Figure 23. Camera viewer settings

- Configure:
  - Resolution e.g., 1920×1080 (FHD)
  - Frame rate e.g., 25 fps

The selected resolution and frame rate are reflected in the viewer header and the displayed stream updates accordingly. See Figure 23 and Figure 24.

Figure 24. Camera viewer with updated resolution and frame rate setting

Note:

If the stream does not start, re-run `setup_hailo_sensor.sh` and restart the camera-viewer-server.

Basic connectivity check can be done by running

`ping 10.0.0.1`

3.3. Object Detection

Object detection can be enabled through the Camera Viewer interface.

Typical capabilities include:

- Real-time detection
- Bounding box overlay
- Class labeling

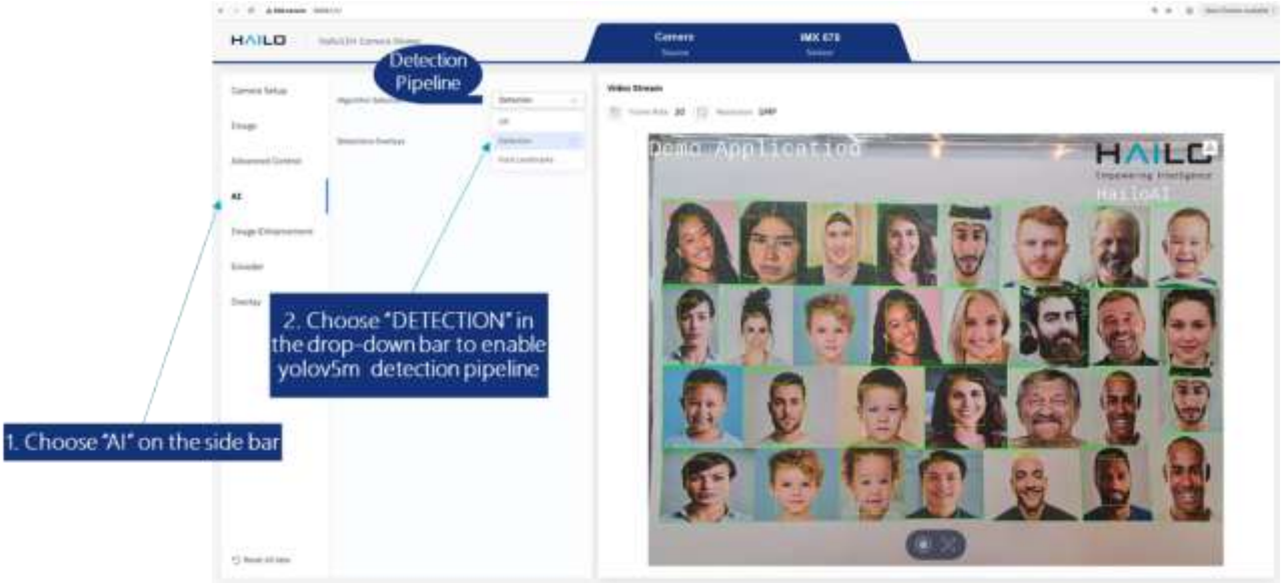


Figure 25. Camera viewer object detection setting

4. Troubleshooting

Problem	Possible Cause	Possible Solution
Hailo-15H SBC is unresponsive after power on	Bad or missing SD card image	• Make sure the DIP switch are set correctly • Re-program the SD card image
No video on the development host when the demo application is running	Video stream is blocked by the firewall	• Check the configured firewall rules. This can be done by using the command: `sudo ufw status`

5. Hailo-15H SBC 2.x Contents

Table 1 below lists the part numbers for the Hailo-15H SBC Kit components.

Table 1. Hailo-15H SBC 2 kit

Hailo Part Number	Part Description	Quantity
HSB2HA1C4GA ES	Hailo-15H SBC 2.x	1
PCBA#HSB15A1C4GA	Hailo-15H PCB 2.x - HSB2HA1C4GA REV 2.0	1
MHS00009	HS-22x22x15.25-Thermal-Pad	1
SDC0001	32GB Sandisk Ultra microSDHC Class10 Card 100MB/s	1
	USB-to-micro-USB cable	1
MECH0006	Screw Round Head Philips M3 Plastic 10.00mm	4
MECH0020	Hex Spacer Threaded M3 Plastic 20.00mm	4
PSU000002	PSU, 100-240VAC C14, 12V/5A 2.1x5.5x11.0mm	1

Please note that the Hailo-15H SBC kit does not include the camera module and cable which should be ordered separately.

5.1. Supported Camera Sensor Modules

The Hailo-15H SBC 2.x supports several camera sensor modules that are fully integrated with the Hailo-15H VPU. These modules utilize a common connector, allowing for direct connection via the FAW-1233-03 cable without the need for an external adaptor board. The modules are available for order as optional accessories from Hailo.

For a complete list of all supported camera sensor modules, please refer to the Hailo-15 Approved Vendor List. This document can be found in the Hailo Developer Zone

The **IMX678 camera sensor module** is the most recommended for use with the Hailo-15H SBC

Note: The camera sensor modules purchased from Hailo include a built-in oscillator, whereas usually, the vendors module datasheets will not include a description of it.

